Qualifying Exam in Microeconomics
Arizona State University
June 2014

There are four questions and you have four hours to complete the exam.

1. Consider a Robinson Crusoe economy. Yams, y, are produced from labor, x, according to the production function $y = x^{1/2}$. Robinson's preferences are given by:

$$U^R(y_R, 1-x) = y_R^{1/2}(1-x)^{1/2}$$

where y_R denotes Robinson's consumption of yams. Let p denote the price of yams. Robinson is the sole owner of the firm.

(a) First define a Walrasian equilibrium as it applies to this economy. You only need to define it.

Second, given prices, and some production plan of the firm, specify precisely Robinson's budget set.

Third, normalize prices so that $p = 1$. Find a Walrasian equilibrium. You must show that the Walrasian equilibrium you propose is indeed a Walrasian equilibrium.

(b) Suddenly a person named Friday arrives to the island where Robinson lives. Friday becomes the owner of the firm and simply consumes all of the yam profits. Assuming that Friday supplies no labor, describe the new Walrasian equilibrium. Compare the equilibrium wages and employment level to that found in (a).

(c) Can the equilibrium wage rate change because of a change in the ownership of the firm? Please *briefly* relate your answer to the solution in (a) and (b).

If CRS not

2. Let the consumption set X be $\mathbb{R}^L_+$ for $L \geq 2$ goods, and let the *at-least-as-good* relation $\succsim \subset X \times X$ be *complete*, *transitive*, *continuous*, *locally nonsatiated*, and satisfy $x \succsim 0$ for every $x \in X$. For $x \in X$ let $C(x) = \{x' \in X \mid x' \succsim x\}$, the no-worse-than-set for x. Define a *minimum wealth function* $E(p, x)$ by

$$E(p, x) = \min_{y \in C(x)} p \cdot y$$

where $p \in \mathbb{R}^L_{++}$ and $x \in X$. Note that the second argument of E is a consumption plan, not a utility number. Let $h(p, x)$ denote the solution correspondence to this expenditure minimization problem and $d(p, w)$ denote the ordinary (Walrasian) demand correspondence:

$$d(p, w) = \{y \in B(p, w) \mid y \succsim x \text{ for every } x \in B(p, w)\}$$

where $B(p, w) = \{x \in X | p \cdot x \leq w\}$ is the budget set at $(p, w) >> 0$. Fix $p >> 0$. Since $\succsim$ is complete, transitive, and continuous, both $d(p, w)$ and $h(p, x)$ are nonempty for every $(x, w) \in X \times \mathbb{R}_+$, facts you may assume in what follows.

You are asked to prove that $E(p, \cdot)$ *represents* $\succsim$ in the following steps. (The representation is called a *money-metric* utility, a phrase you have heard before.)

(a) If $x'' \succsim x'$, then $E(p, x'') \geq E(p, x')$.

(b) If $y \in d(p, E(p, x))$, then $y \sim x$. (Suggestion: to prove that $x \succsim y$, consider the cases $y = 0$ and $y \neq 0$ separately. If you cannot prove part (b) in its entirely, then *assume* it for part (c).)

(c) If $E(p, x'') \geq E(p, x')$ for $x', x'' \in X$, then $x'' \succsim x'$.

Redo!!

3. A risk-neutral monopoly firm sells insurance to a single, risk-averse consumer. The consumer has initial wealth of $w > 0$ and faces a potential loss of size L with chance p, where $0 < L < w$ and $0 < p < 1$. The loss chance takes on one of two values, p_1 or p_2, where $0 < p_1 < p_2 < 1$. The consumer knows whether $p = p_1$ or $p = p_2$ but the firm does not. The firm believes that $p = p_2$ with probability $\lambda \in (0, 1)$. Call a consumer with loss chance p_i a *type-i consumer.*

The consumer has expected utility preferences represented by a *strictly concave* vN-M utility u, which is C^1 with $u' > 0$ on its domain of $\mathbb{R}_+$. The firm offers the consumer a menu of contracts $\{(x_i, t_i)\}_{i\in\{1,2\}}$ where t_i is the transfer that a type i consumer pays to firm in *both* states of the world and x_i is the money that the firm pays to a type i consumer if the loss occurs, where the contract is constrained to satisfy $(0, 0) \leq (x_i, t_i) \leq (L, w)$ for $i = 1, 2$. The firm's expected profit from a contract (x_i, t_i) bought by a type-i consumer is $t_i - p_i x_i$. Define a function U by

$$U(x, t, p) = pu(w - L + x - t) + (1 - p)u(w - t)$$

for $p \in \{p_1, p_2\}$ and $(0, 0) \leq (x, t) \leq (L, w)$, the expected utility of a consumer with loss chance p from a contract (x, t).

The firm chooses a menu of contracts $\{(x_i, t_i)\}_{i\in\{1,2\}}$ to maximize expected profit subject to the "incentive compatibility" constraint that each consumer type i weakly prefers (x_i, t_i) to (x_j, t_j), $j \neq i$; and the "participation" constraint that each type weakly prefers its contract to the no-trade contract, $(0, 0)$. The firm knows w, L, p_1, p_2, λ, and $u(\cdot)$ (but to repeat not which type the consumer is).

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(a) Write down the firm's constrained profit-maximization problem.

(b) Does it follow from the given assumptions that U satisfies the *strict single crossing property* (SSCP) in p and (x, t) on its domain?[1] If so, prove it; if not add conditions so that it does and prove your assertion. No matter what, *assume* in what follows that U satisfies the SSCP in p and (x, t).

(c) Argue that any incentive-compatible menu must be $\geq$-ordered, that is, either $(x_1, t_1) \geq (x_2, t_2)$ or $(x_2, t_2) \geq (x_1, t_1)$. (This should take about a sentence. If you can't see it, *assume* it in what follows.)

(d) Use the SSCP and part (c) to prove that any solution to the firm's problem is *monotone*: $(x_2, t_2) \geq (x_1, t_1)$.

(e) Prove that at any solution to the firm's problem the type-2 consumer gets full coverage: $x_2 = L$. (Suggestion: don't use calculus.)

more likely to loss

$(x_1 \geq x_2 \quad t_1 \geq t_2)$

or

$(x_1 \leq x_2 \quad t_1 \leq t_2)$

$(x_1 < x_2$ or $t_1 < t_2)$

and $(x_1 > x_2$ or $t_1 > t_2)$

[1] The partial order on the types is the usual inequality $>$ on the reals and on contracts it is the usual vector inequality $>$ on $\mathbb{R}^2_+$ ($z > y$ means $z \geq y$ and $z \neq y$).

4. A potential worker can select any effort level $e \in [\underline{e}, \overline{e}] \subseteq [0,1]$. For any given wage, w, and effort level, e, the worker's utility is $v(w) - e$ where $v'(w) > 0$, $v''(w) < 0$. If the worker chooses not to accept the job, his reservation utility is $\bar{u}$. Exerting an effort level e results in random profits $\pi \in \{\pi_L, \pi_H\}, \pi_L < \pi_H$ with distribution $P(\pi_L|e) = 1 - e$. The employer is risk neutral.

   (a) Suppose that the firm *observes* the effort level selected by the agent and conditions monetary payments accordingly. Characterize the minimum cost of achieving any given effort level e and the profit maximizing level.

   (b) Show that the profit-maximizing level of effort found above, remains unchanged when π_L and π_H each rise by the same amount x. How does the profit-maximizing level of effort change when π_L falls by x but π_H rises by x at the same time?

   (c) Show that $e' > e$, implies that the distribution of profits when e' is selected first order stochastically dominates the distribution when e is selected and that the two corresponding conditional probability densities satisfy the monotone likelihood ratio condition.

   (d) Suppose now that effort level is *not observable* by the principal (or that payments cannot be contracted on effort) but that payments to the worker can depend on π. Show that if payments can only depend on realized profits, any profit-maximizing effort that can be induced is either the highest effort level $\overline{e}$ or the lowest effort level $\underline{e}$.